Robust self-supervised learning across clinics

Research proposal, 3-year PhD · Tania Rahman · Sample draft

1. QUESTION

How can self-supervised models trained on small medical image sets stay accurate when images come from new scanners and clinics?

2. BACKGROUND

Contrastive pre-training helps when labels are scarce, yet reported gains are usually measured within one site. Three strands matter: (a) label-efficient pre-training; (b) domain shift in medical imaging; (c) evaluation that separates site effects from disease signal. The host group's recent paper names scanner shift as an open question.

3. METHOD

- Pool unlabelled images from several public sources and pre-train with a contrastive objective plus a site-invariance term.
- Evaluate on held-out sites with fixed splits, reporting per-site scores, not only averages.
- Ablate each component and report negative results.

4. PLAN

Year	Work	Output
1	Reproduce baselines; build the cross-site benchmark	Benchmark and workshop paper
2	Develop and test the robustness method	Conference paper
3	Clinical collaboration study; write thesis	Journal paper and thesis

5. WHY THIS GROUP

Access to multi-site data, prior work on exactly this problem and a record of supervising students to publication.

6. RISKS

Public data may lack enough site variety. Mitigation: use the group's partner data once agreed, and publish the benchmark either way.

SAMPLE